

CALCULUS 1

November/December 2013

Time : 2 hours

Candidates may attempt ALL questions in Section A and at most TWO questions in Section B. Each question should start on a fresh page.

SECTION A (40 marks)

Candidates may attempt ALL questions being careful to number them A1 to A4.

A1. (a) Solve the following inequality $\frac{2x+3}{x-5} > 3$.

(b) Show that $|x+y| \leq |x| + |y|$ for all $x, y \in \mathbb{R}$.

[4,4]

A2. (a) Define what is meant by saying that l is the **limit** of a sequence u_n .

[2]

(b) Evaluate each of the following limits.

(i) $\lim_{n \rightarrow \infty} (\sqrt{n+8} - \sqrt{n})$, (ii) $\lim_{n \rightarrow \infty} \frac{5n^4 - 7n^2 + 5}{9n^4 + 3n^3 + n^2 + 1}$.

[3,3]

A3. (a) Estimate the value of $\sqrt[5]{36}$ using differentials.

[4]

(b) Use the Mean Value Theorem to prove that $\sqrt{1+x} < 1 + \frac{1}{2}x$ for $x > 0$.

[5]

A4. (a) Show that $\int_0^1 x^2 \tan^{-1} x \, dx = \frac{\pi}{12} - \frac{1}{6} + \frac{1}{6} \ln 2$.

[6]

(b) Find each of the following integrals.

(i) $\int \cos^5 x \, dx$ (ii) $\int \frac{dx}{1 + \sin x}$.

[5,4]

SECTION B (60 marks)

Candidates may attempt TWO questions being careful to number them B5 to B8.

- B5.** (a) (i) Prove that if $0 < a < b$, then $a^2 < b^2$. [4]
 (ii) If $a^2 < b^2$, does it necessarily imply that $a < b$? If so, prove it and if not give an example that illustrates your answer. [4]
 (b) (i) Define in terms of ε and δ , what is meant by saying that a function $f(x)$ is continuous at a point $x = x_0$. [2]
 (ii) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} x^2 + 1, & x \geq 2 \\ ax + b, & -1 < x < 2 \\ 5 - 2x, & x \leq -1, \end{cases}$$

where a and b are constants. If f is continuous on $\mathbb{R}$, find the values of a and b . [6]

- (c) (i) State the Mean Value theorem. [4]
 (ii) Use the Mean Value Theorem or otherwise to prove that

$$1 - \frac{a}{b} < \ln \left(\frac{b}{a} \right) < \frac{b}{a} - 1$$

if $0 < a < b$. [6]

- (iii) Hence or otherwise show that

$$\frac{1}{6} < \ln(1.2) < \frac{1}{5}.$$

[4]

- B6.** (a) (i) Define what is meant by $\lim_{n \rightarrow \infty} u_n = -\infty$.
 (ii) Show using the definition, that $\lim_{n \rightarrow \infty} (1 - 2n) = -\infty$.
 (iii) Prove that $\lim_{n \rightarrow \infty} \frac{c}{n^p} = 0$, where $c \neq 0, p > 0$. [3,4,4]
 (b) (i) What is the relationship between differentiability and continuity?
 (ii) Hence prove this relationship. [3,6]
 (c) (i) If $e^{xy} + y \ln x = \sin 2x$, find $\frac{dy}{dx}$.
 (ii) If $y = x^x, x > 0$, find $\frac{dy}{dx}$. [6,4]

- B7.** (a) Evaluate each of the following limits.

(i) $\lim_{x \rightarrow 0} \left(\frac{1 + 3x}{\sin x} - \frac{1}{x} \right)$ (ii) $\lim_{x \rightarrow 0^+} x^x$. [5,4]

- (b) (i) Prove that if $f(x)$ is continuous on $[a, b]$, differentiable on (a, b) and if $f'(x) \geq 0$ for every x in (a, b) , then $f(x)$ is monotonic increasing on $[a, b]$.
(ii) Verify Rolle's Theorem for $f(x) = x^2(1 - x)^2$, $0 \leq x \leq 1$. [5,4]
- (c) Sketch the graph of $f(x) = \frac{x-1}{x+2}$ indicating any asymptotes and any points of intersection with the axes. [10]

B8. (a) Evaluate each of the following integrals.

(i) $\int \frac{\sqrt{x}}{\sqrt{x}+1} dx$ (ii) $\int \sin^{-1} x dx$. [5,5]

- (b) Find the area of the region bounded by the graph of $f(x) = 1 - x^2$, $x = 0$, $x = 1$ and the x -axis by calculating the limit of the Riemann Sums. [10]

- (c) Find

$$\int \frac{dx}{(x^2 + 1)(x + 1)}.$$

[10]

END OF QUESTION PAPER